Mikk **Kruusalu**

Machine learning · Physics · Numerical modelling

+37253030327 kruusalu.mikk@gmail.com https://mikruu.ee mikk-kruusalu Mikk Kruusalu

About

Radar development team lead with engineering experience in control loop design, machine learning, embedded firmware and electronics. I have MSc in Applied Physics and Data Science and I am passionate about modelling physical systems.

Work experience

Radar Development Team Lead

CAFA Tech — July 2026 - Present

Team lead for developing a cost-effective X-band radar for small drone detection.

- Signal processing.
- High-level radar architecture modelling.
- Efficient communication between mechanical, RF, software engineers and academics.
- Mentoring.

Development Engineer

CAFA Tech — Mar 2023 - June 2026

- Custom visual inertial odometry system for position hold of a multirotor in GPS-denied environments.
- Drone tether system control loop firmware and electornics.
- Energy Team Lead in developing a small-scale microgrid.
- Machine learning model for classifying objects based on their trajectories.
- Developed a Data Acquisition System based on Beckhoff PLC and Timescale database.
- Rapid prototyping of heat engines.
- Developed a battery pack control system PCB and firmware.

Robot Technician

Starship Technologies — June 2022 - Feb 2023

- Repaired mechanics and electronics including PCBs
- Create tools to improve the robot's repair process

Skills

Languages English (C1), Estonian

Programming **Python** (Jax, Pytorch, Scipy, Numpy, Scikit-learn, Sktime), Rust, Git, Julia, R, C++, Matlab

Programs OpenFOAM, ElmerFEM, Paraview, Autodesk Fusion 360, Kicad

Education

MSc Applied Physics and Data Science

TalTech, Cum laude — Sep 2024 - June 2026

- Mathematical Modelling, Machine learning, Deep Learning, Numerical methods, Scientific computing

Erasmus exchange student

Denmark University of Techonolgy — Sep 2023 - Jan 2024

- Dynamical Systems, R, Deep learning, State Space Models

BSc Applied Physics

TalTech, Cum laude — Sep 2021 - June 2024

- Thesis – modelling nerve signal propagation in an axon
- Mathematical modelling, Physics, Probability and Statistics

Projects

Visual-inertial odometry

Master's thesis — Read more here!

Developed a custom VIO algorithm for position hold of a multirotor drone operating at high altitude low visibility conditions.

Implementing PaiNN

Course project in DTU — MrCogito/Deep_learning_2023

State of the art graph neural network architecture for predicting physical properties of molecules being much faster than conventional methods.

Robotex

- Competed in line following, water rally, maze solving, folkrace. I have won also a few prizes.
- Building robots involved 3D modelling, custom PCBs, also algorithmic thinking and programming in C.

Deep learning

Course project — mikk-kruusalu/deep_learning_project

Implementations of different architectures such as CNN, RNN, GAN, VAE, PINN for various model problems.

Nerve signal modelling

Bachelor thesis — mikk-kruusalu/heimburg_jackson

I used Jax for analysing nonlinear dispersive wave phenomena in nerve axons. Learned about the electrophysiology of nerve signals.

Graph topology optimisation

Course project — https://mikruu.ee/assets/power-grid-optimisation.pdf

I analysed what is the optimal topology for a small scale deployable power grid to be tolerant for node attacks while delivering the most power.